$$a_x = -g\left(\frac{-2\rho(a_x)}{C\rho A}\right)\left(\frac{C\rho A}{2\rho g}\right)$$

$$a_x = a_x$$

$$\Rightarrow \frac{dV_x}{dt} = -g\frac{(V_x)^2}{(V_t)^2}$$

$$\int \frac{1}{(V_x)^2}\,dV_x = \int \frac{-g}{(V_t)^2}\,dt$$

$$-\frac{1}{V_x} = \frac{gt}{(V_t)^2} + C$$

$$\frac{1}{V_x} = -\frac{gt + C(V_t)^2}{(V_t)^2}$$

$$V_x = \frac{(V_t)^2}{C(V_t)^2 - gt}$$

sub $t=0$, $V_x = V_{ix}$

$$V_{ix} = \frac{(V_t)^2}{C(V_t)^2}$$

$$C = \frac{1}{V_{ix}}$$

$$V_x = \frac{(V_t)^2}{\frac{(V_t)^2}{V_{ix}} - gt}$$

$$= \frac{V_{ix}(V_t)^2}{(V_t)^2 - gtV_{ix}}$$

$$V_x = \frac{V_{ix}(V_t)^2}{(V_t)^2 - gtV_{ix}}$$

$$\frac{dS_x}{dt} = \frac{V_{ix}(V_t)^2}{(V_t)^2 - gtV_{ix}}$$

$$S_x = V_{ix}(V_t)^2 \int \frac{1}{(V_t)^2 - gtV_{ix}}\, dt$$

$$= -\frac{V_{ix}(V_t)^2}{gV_{ix}} \ln\left|(V_t)^2 - gtV_{ix}\right| + C$$

when $t = 0$, $S_x = 0$

$$0 = -\frac{(V_t)^2}{g} \ln\left|(V_t)^2\right| + C$$

$$C = \frac{(V_t)^2}{g} \ln(V_t)^2$$